

Co-designing an Integrated Digital Education Portal for the Eastern Cape Rural Learners

Lutho Makasi*
Student, Undergraduate, Walter
Sisulu University
luthomakasi1@gmail.com

Yolanda Gwala
Student, Undergraduate, Walter
Sisulu University
yolandagwala02@gmail.com

Anam Baza
Student, Undergraduate, Walter
Sisulu University
bazaanam@gmail.com

Andile M Lwanga
Student, Undergraduate, Walter
Sisulu University
andilemlwanga@gmail.com

Chuma Makalima
Student, Undergraduate, Walter
Sisulu University
chumamakalima8@gmail.com

Supervisors/Mentors
Nobert Jere and Nosipho Mavuso,
Lecturers, Walter Sisulu University

ABSTRACT

The digital divide gap remains widening as poor and rural communities continue to struggle to access and benefit from emerging technologies. However, access to digital technology has bridged physical and economic barriers in some communities by allowing access to information, which, in turn, has pushed the digital economy or Industry 4.0 (4IR) forward. Unfortunately, in African rural communities, access to digital technology is still distant. Research on rural schools indicates that the greatest portion of those living in rural areas access their internet through their mobile phones unlike in urban areas where the internet is accessed through wireless and digital subscriber lines (DSL) in addition to mobile phones. As a result, these differences have created a digital divide between urban and rural areas as well as among provinces in South Africa. Additionally, rural schools produce poor results and have poor ICT infrastructure and physical resources, and the economic status of the learners is equally a huge worry. Rural schools have challenges in accessing books and the latest learning material, affecting the quality of education. Most critically, most digital education solutions are designed and developed without the involvement of rural learners and teachers. In this study, the Eastern Cape schools in South Africa are considered. Through participatory design, including learners, teachers and key education stakeholders, a digital education portal is co-designed. Results have shown that education stakeholders are willing to participate and contribute to digital education platforms. Stakeholders would like a simple design that considers their views and input in order to easily use and benefit from the designed platform.

CCS CONCEPTS

• Human Computer Interaction; • Co-Design; participatory design; • Sustainable Development;

KEYWORDS

Digital Education, Quality Education, User Interface Design, Co-Design

ACM Reference Format:

Lutho Makasi*, Yolanda Gwala, Anam Baza, Andile M Lwanga, Chuma Makalima, and Supervisors/Mentors. 2023. Co-designing an Integrated Digital Education Portal for the Eastern Cape Rural Learners. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3544549.3583839>

1 INTRODUCTION

In many Sub-Saharan African countries, the learning gap between urban and rural areas is a recurrent issue. Previous studies have found that the differences in student characteristics and school resources between urban and rural areas are what causes the rural-urban learning gap. Rural students' learning performance in many Sub-Saharan African countries is poorer than that of urban students, which is seen as a concerning gap related to knowledge disparities or education inequality. The achievement gap between urban and rural students is a global phenomenon [1]. In sub-Saharan Africa, rural children perform consistently more poorly than urban children, and compared with the world average, the gap is particularly large [2]. Urban schools have more learning resources and good infrastructure compared to rural schools. Most urban schools have toilets, whereas rural schools do not always have appropriate toilets [3]. The lack of adequate funding from the government for rural schools continues to be an issue as it puts operations and developments of these schools on halt. Often, teachers in rural schools are expected to perform duties outside of their profession which they are not financially compensated. This triggers the rural schools having more underqualified teachers and the urban schools with more qualified teachers. Qualified teachers prefer to teach in urban schools as there is better access to water, electricity, health services, good infrastructure and for leisure.

This study focuses on a solution for the phenomenon, targeting the rural schools in the Eastern Cape, province of South Africa. The Eastern Cape province in South Africa faces significant disparities in its education system, largely due to poverty and unequal access to resources. This has resulted in a significant disparity in educational achievement between fee-paying and non-fee-paying schools, with the latter achieving a much lower pass rate of 64.9% compared to

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '23, April 23–28, 2023, Hamburg, Germany
© 2023 Copyright held by the owner/author(s).
ACM ISBN 978-1-4503-9422-2/23/04.
<https://doi.org/10.1145/3544549.3583839>

the 93.7% pass rate of the former. The persistent challenges in the education system have highlighted the need for intervention to provide better learning opportunities and support student success. Over 70% of learners in the Eastern Cape attend quintile 1, 2 and 3 schools, which are supposed to receive more support but instead face inadequate conditions, including a shortage of study materials, poor infrastructure and overcrowding in classrooms. These challenges negatively impact student performance and are often ignored in the efforts to support underperforming schools, highlighting the need for intervention to provide better learning opportunities and support student success.

There is a significant disparity in educational outcomes between male and female students, with only 7% of women pursuing careers in STEM fields [9]. To address these challenges, we saw a clear need for both a hands-on community approach and a digital solution that can provide equal access to quality education while also engaging students in the design process through participatory design methods. The study was undertaken to address the United Nations Sustainable Development Goal 4: Provide Quality Education.

This study focuses on a solution for the phenomenon, targeting the rural schools in the Eastern Cape, province of South Africa. The solution developed is “Makukhanye”, an isiXhosa word that means “to Bring Light”. The Makukhanye platform aims to incorporate the ideas and preferences of students in the Eastern Cape by engaging them in the design process. Through participatory design methods, students inform the best times, techniques and methods for learning both at a human level and in a digital infrastructure approach. This user-centred approach to design helps ensure that the final product meets the specific needs and requirements of students in the region. By empowering students to shape their education, the project promotes a more effective and efficient learning experience, providing greater access to quality education for all students in the Eastern Cape province.

This submission is to present the idea and solution that the authors have for the Students Design Competition during the CHI 2023 conference. The following literature has been considered to assist in understanding the current state of education in rural areas.

2 RELATED LITERATURE

Disadvantaged communities lack the obligatory physical resources and crucial services for sanitation, water, roads, transportation, power and information and communication technology [15]. This deprivation also affects rural schools in the sense that they lack financial backing, which makes both teaching and learning more complicated. Poverty is also widespread in many disadvantaged communities, which leads to a severe threat to the provision of high-quality education and awareness of current trends like 4IR. These problems stem from a diversity of features, including internal school classifications and peripheral environments, which involve local communities and educational specialists [16].

In most communities, there is a high need for the enhancement of infrastructure and structures that convey exposure to places that are outliers, especially disadvantaged communities [17]. This entails several sectors comprising the government to advance wifi access, improve attention through access to the internet as well as making sure that in most places in the state, the general public

that is living there has admission or access to cell phone coverage and internet exposure to mention a few. It is also clear that data is expensive in most communities or nations, meaning to address digital inequalities in disadvantaged communities, there is a prerequisite to go over or review how individuals from those communities survive or live, of which most of them are subsistence farmers, and this makes them vulnerable to their economic status.

Therefore, the government and technologists have a momentous role when it comes to reviewing the price of data since most citizens do not have to salvage the situation. This reveals that if the government can either subsidise or safeguard that the private suppliers of data and the internet make their services and products affordable, this will certainly help in addressing digital inequality in the disadvantaged communities [18].

Furthermore, it is a fact that access to progressive technology in Africa is held in reserve by infrastructure restrictions corresponding to the absence of electricity and little tele-density, internet compactness and broadband diffusion [19]. More so, technological blockages include the shortage of reliable use programme design interfaces and common data languages for the improved fusion of basically self-sufficient methods and also exposure to the fears of cyber-attacks. As a consequence, fast-tracking the physical connectivity of fibre-optic networks as well as virtual display places are important for the progress of technology in the region, as this will influence and lessen unit rates for the disadvantaged, especially in rural communities.

According to [4], the Minister of Basic Education in South Africa, Angie Motshekga, reportedly said that most schools in the Eastern Cape fail to meet the minimum norms and standards set under the South African Schools Act. Only 1% of schools in the province have adequate fencing which protects learners, teachers and classrooms against violence and vandalism. The province also accounts for 45% of schools in South Africa that rely solely on pit latrine toilets. Making matters worse, vital school infrastructure projects have been halted due to a lack of funding. Out of South Africa's 23,259 operational public schools, 5,290 – roughly 23% – are in the Eastern Cape. But a dire lack of resources, isolated classrooms and the mismanagement of funds intended to improve the level of education in the province continue to plague the pupils [5]. It also indicates that the Eastern Cape province has the second highest percentage of learners in the country that did not pay school fees (81%). This is significantly higher than the national average (62.4%). In 2013, 87.2% of the learners attended schools with nutrition programmes, and 85.1% of them made use of the service.

The Department of Education statistics for a couple of years show that Eastern Cape is number 7 out of 9 provinces in pass rate stats for matric 2021, with a 73% pass rate. Quintile 1 schools (non-fee-paying schools) achieved a 64.9% pass and quintile 5 schools (fee-paying schools) achieved a 93.7% [6]. This shows the difference in the quality of education between fee-paying schools and non-fee-paying schools. The national and poverty breakdown on quintiles in the national poverty table for 2014 given by the Minister of Education of Western Cape, Donald Grant in a media release stated that the Eastern Cape province has 27.3% schools on quintile 1, 24.7% schools on quintile 2, 19.6% on quintile 3, 17% on quintile 4 and 11.4% schools on quintile 5. Given the Eastern Cape schools quintile breakdown, the quintile 1, 2 and 3 schools are meant to

receive more financial support from the province's department of education, yet these are the schools that suffer more than the quintile 4 and 5 schools. The infrastructure and textbook delivery in the quintile 1, 2 and 3 schools continue to get worse each year, with these schools catering for more than 70% of the learners in the province [7].

According to [8], parents, teachers and activists took the Eastern Cape education department to court for its failure to deliver books and stationery to over 3,000 schools in the province. In September 2022, the provincial department was yet to deliver stationery and textbooks to every school, blaming “unprecedented budget short-falls”. This just shows one of the reasons the quintile 1 schools are not particularly doing well academically.

This literature section presents the challenges, initiatives and some of the documented efforts in different areas to improve education. There are challenges that affect high quality education in rural schools. Section 3 covers a summary of the overall methodology that we considered for this study.

3 RESEARCH APPROACH

The research team utilised a participatory design approach and user-centred design methods to address the limited access to quality education in the Eastern Cape province. Through engaging with 26 students at two secondary schools, the team gained a comprehensive understanding of the students' needs and challenges. These insights were incorporated into the design of a digital learning portal, which was developed through a combination of questionnaires, mock-up presentations and prototypes. The team also conducted a review of existing solutions to inform the design of the portal, ensuring that it was tailored to the specific needs of the students in the Eastern Cape province. The result was a solution that addressed the root causes of the problem and improved access to quality education in the region.

Design process

- **Problem definition and research:** This phase involved defining the problem, conducting a review of existing solutions and conducting user research to gain a deeper understanding of the users' needs and preferences.
- **The design process for the digital learning portal** followed a user-centered approach to create a solution that meets the needs and preferences of the target users. The design process consisted of the following elements:
- **Ideation:** During this phase, the team generated ideas and concepts for the solution, taking into consideration the insights gathered from the research phase.
- **Prototyping:** The team used low-fidelity prototyping methods such as sticky notes and paper sketches to create a rough draft of the design. This phase also involved creating a high-fidelity digital prototype using Adobe XD.
- **User testing:** The team conducted user testing to gather feedback from the target users on the design and make necessary improvements.
- **Implementation:** Based on the feedback from the user testing phase, the team made any necessary changes to the design and then proceeded to implement the solution.

- **Evaluation:** The final step involved evaluating the solution and making any necessary improvements based on user feedback.

This design process followed a user-centred approach, with a focus on co-design and collaboration between the design team and the users. The process ensured that the solution met the needs and preferences of the target users, leading to a successful and effective digital learning portal.

4 UNDERSTANDING OF THE CONTEXT

Upon receipt of research clearance from Walter Sisulu University, the team conducted research and design of the platform over 5 days at each of the schools, guided by the San Code of Research Ethics and the Global Code of Conduct at two schools in Mdantsane, Eastern Cape, the Vulamazibuko High School and the Masixole Secondary School. Firstly, we introduced the team, our mission and vision, and the main reason we came to the school to engage with them. To see if they understood the concept, we asked them a few questions to help them get on board. We used a questionnaire to gather data from the students.

Our findings from the primary research we conducted with 26 students were that: There were about 50-60 students in one classroom, and 40-60% of the time, the students failed to concentrate due to rowdiness in the classroom. Only 10 learners had 4 of the 7 expected textbooks, 9 were sharing textbooks in groups of 3, 2 had 5 of the expected 7 textbooks, and 5 had no textbooks for all subjects for the academic year of 2022. They all stated that they had not managed to complete the syllabus for the past 2 academic years. They did not have sufficient study material to help them prepare for tests and examinations. They felt that the teachers did not give them individual attention to ensure a thorough understanding of subject. They all stated that they have smart cellphones. Only 4 of the 26 had exposure to at least one digital learning platform. They expressed interest in a digital learning platform that be ease to navigate.

A further survey was conducted with more high school pupils of the Eastern Cape province to identify the best mode to provide the solution, and the results showed that 89% had smartphones, and 36% had a reliable internet connection.

A shortage of study material, poverty, poor infrastructure, overcrowding and gender inequality are some adverse conditions plaguing the Eastern Cape schools. While these conditions have dire consequences on the performance of teachers and pupils, they are hardly considered when looking into ways to support underperforming schools. Despite improvements in the South African Department of Basic Education system, deep inequalities continue in the Eastern Cape education system. Overcrowding compromises the education of learners in the Eastern Cape schools, “We unfortunately can't accommodate all learners; they have to rotate and come for 3 or 4 days a week, and we can't finish the syllabus”, said Yandisa Ndala, a teacher at Mncwathi secondary school in the Eastern Cape talking to Nkululeko Nyembezi a journalist at SABC News in December 2022 [10]. Ndala further mentioned that on rainy days, pupils have to go home due to the classrooms with broken windows and leaking roofs, resulting in forfeiting a teaching and learning day, and this being one of the many causes of the syllabus not being

finished [10]. The matriculants in South Africa write a national paper, meaning all matriculants in the country are examined on the same syllabus, and the matriculants from the Eastern Cape are forever disadvantaged compared to the others.

5 CO-DESIGNING THE SOLUTION

At the two secondary schools, the approach of participatory design and co-creation and user centred design methods was taken with a group of 26 learners. We started by showing them templates and platforms that already exist for the exposure/experience of using an E-learning platform. This made it easier for them to note what already exists and what does not exist. They had the chance to share their input and also voice their opinions about the relevance of our platform in relation to their learning experiences. A mock-up of the website was then created using Adobe XD, and sticky notes were facilitated and guided by the team members. The learners discussed with the team why they had designed the mock-up the way they did, highlighting the need for simplified tutorials to be able to better understand the content taught in class and a summary of important notes in the textbook (study slides). Listening intently to the learners, the team took valuable insights into the learners' needs which guided the team in the further development of the application [13, 14].

6 VISUAL CUES FOR THE DESIGN PROCESS

The visual cues for the design process are an important aspect in ensuring the effectiveness and efficiency of the design process. These cues serve as visual indicators of the progress made in the design process and help to ensure that the team stays on track with the design goals and objectives. In the design process, visual cues can come in various forms, such as sketches, wireframes, prototypes and visual representations of user flow. These visual cues can also help to communicate the design ideas effectively to the team and stakeholders, making it easier to understand and make decisions. Furthermore, incorporating visual cues in the design process can also help to identify and address potential design challenges and limitations early on, ensuring that the final product is of high quality and meets the needs of the target audience.

7 PORTAL FEATURES

The team developing the student portal for high school students in rural areas leveraged several features commonly found in learning management systems (LMS) to create a solution that meets the unique needs of this population. Some of the features that were used include:

1. **Digital Libraries:** The portal offers students access to a wealth of educational materials, including online books, articles and other resources. This feature provides students with a wealth of information they can use to learn and explore different subjects.
2. **Online Teaching:** The portal also includes online teaching components, allowing students to access educational videos, lectures and tutorials from anywhere, at any time.
3. **Academic Support:** The portal includes messaging features that allow students to connect with tutors for additional support and guidance.

4. **Practice Questions:** The portal includes interactive practice questions, allowing students to test their understanding and receive immediate feedback.

By combining these features in a single platform, the team was able to create a comprehensive and user-friendly platform that meets the unique needs of high school students in rural areas. The goal was to provide these students with the resources they needed to succeed academically, regardless of where they lived or their socio-economic status.

8 TECHNOLOGY AND TOOLS USED

In the development of the digital learning management system, the team utilised various aspects of user interface design and prototyping tools. These tools helped in creating a user-centered design that was centred around the needs of the students and their learning experiences. The team used Adobe XD to create mock-ups of the website, which were then tested and refined based on feedback from the students. The use of sticky notes also allowed for an iterative design process, where the team could quickly make changes and test new ideas. By using these technologies, the team was able to ensure that the final product was not only functional but also visually appealing and user-friendly, meeting the needs and expectations of the intended users.

9 IMPLEMENTATION

The design process consisted of several key elements to ensure its success and alignment with ACM standards. The team first applied a participatory design approach and user-centred design methods to engage with a group of 26 students at two secondary schools in the Eastern Cape. This helped to understand the students' needs and preferences in terms of online learning and academic support. Then, the team conducted a competitor analysis to understand the current offerings in the market and identify any gaps.

The team then developed personas based on the insights gathered from student engagement and competitor analysis. This helped to ensure that the design solution was tailored to the specific needs and requirements of the target audience. Besides, the team created a low-fidelity and high-fidelity design mock-up, guided by the students' discussions and needs and used visual cues to track progress throughout the design process.

10 COMPETITIVE ANALYSIS

The issue of limited access to quality education in the Eastern Cape province has prompted various initiatives to improve the education system in the region, but these efforts have been insufficient to address the problem. While technological solutions such as e-learning platforms and digital libraries have been proposed, they have not been widely adopted due to challenges like limited internet access and technical expertise. A thorough review of existing solutions has revealed the need for a more user-centred approach that prioritises the needs of learners and maximises the use of available technology and infrastructure. To achieve this, it is important to consider the unique challenges and opportunities of the education system in South Africa and develop a sustainable and effective solution that meets the needs of all stakeholders.

The co-design approach is critical in addressing limited access to education in the Eastern Cape province. By incorporating the needs and perspectives of students and staff, the solution can better meet their needs, foster student ownership of learning and ensure inclusivity. It offers a promising strategy for improving access to quality education in the region.

The team conducted a thorough analysis of the existing digital learning platforms and evaluated their strengths and weaknesses. This information was used to identify areas for improvement and to differentiate the proposed solution from the existing ones. The team analysed the factors such as user experience, functionality, cost, accessibility and quality of content.

The competitor analysis allowed the team to identify gaps in the market and to create a unique solution that meets the specific needs of schools in Eastern Cape. The findings of the analysis were used to inform the design of the digital learning portal, ensuring that it offers a competitive advantage and addresses the needs of the target audience. The competitor analysis also provided valuable insights into the expectations of the users, which were used to guide the development of the application's features and functionality.

In alignment with ACM standards, the competitor analysis was conducted ethically and transparently, considering the privacy and security of the data and ensuring that the findings were used to improve the proposed solution and not to imitate existing solutions.



Figure 1: Pictures of team co-designing with students

11 DESCRIPTION OF DESIGN SOLUTION

Our solution is to design a digital learning portal for the Eastern Cape schools that focus on online deliveries such as digital libraries, digital online teaching and academic advice to mention a few. We wanted this portal to mainly focus on the quintile 1, 2 and 3 schools because these schools are not doing well academically. This portal has free or affordable online content for all high school students.

For phase 1, the students wanted their portal to mainly focus on the following:

- Simplified notes.
- Understandable video lessons.
- Being able to communicate through messages with the tutors.
- Practice questions.

12 PERSONAS AND SCENARIOS

12.1 Co-Design/participatory design

We used the co-design/participatory design method as referenced in subheading 5: co-designing the solution.

12.2 User Centred Design Method (Persona-based)

We used a persona-based user interface design method. This method starts with deciding on a target market and doing primary research to determine the aspects that will influence our design based on that research. Important information from that research is the main points and wishes the target audience may have.

Based on the concerns raised by the group we spoke to, we came up with a persona on which to base our design. The purpose of this character is to create an idea of our target market and generate a few scenarios that could potentially arise.

12.3 Persona

Uviwe Hlatshwayo

Grade 11 Student

Living Standard Measure: 4 (Least access to wealth)

Masixole Secondary School

Pain Points:

- Too many students in class.
- Teachers being absent for long periods affects the syllabus.
- Textbooks delivered late.
- Teachers not being engaged in the student's learning experience.

Wishes:

- Digital learning platform.
- Opportunity to get extra notes or study material.
- A way for teachers to be more involved.

12.3.1 Scenario 1. On Monday before a big test on Friday, Uviwe is struggling to understand the work in his maths textbook. His teacher does not have time to explain much detail, so she advised him to log into the Makukhanye portal and look for notes on the work that had been covered in the last two weeks. Uviwe gets a different perspective and understands the work a lot better. He also answers extra questions on mock tests that had been uploaded for practice.

Problems solved:

- Limited resources mean many teachers cannot give personal attention; Uviwe was able to get a better understanding of the content.
- Students not fully understanding the work straight from the textbook can get a different perspective.

- Teachers wanting to give extra resources have a financially and environmentally sustainable way of giving extra practice work. Regarding a shortage of study material, teachers and tutors can share videos, simplified notes and articles about certain topics to bridge the gap.
- Because of overcrowding, teachers who want to help are not able to this, however the proposed portal will give them a platform to reach out to many students.

13 DESIGN PRINCIPLES, VISUAL ANALYSIS AND COMPETITIVE ANALYSIS

Visual analysis is a critical evaluation of a design or work of art that measures to what extent the designer used the elements of art and principles of design to create an interesting and effective composition [11].

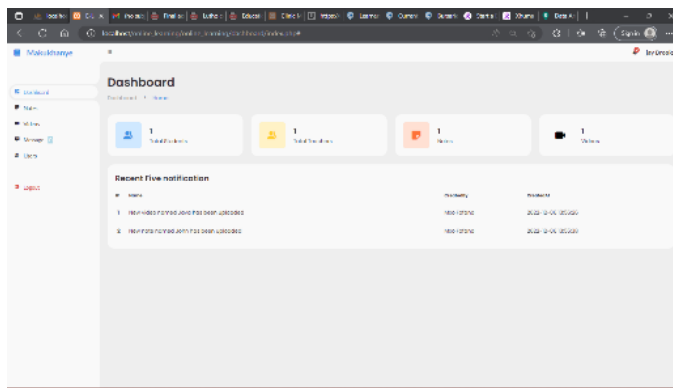


Figure 2: Makukhanye Edu-care dashboard

Space & Proportions – We decided to have the information spaced out as possible while maintaining proportions to make sure each element is easy to navigate. Because there is a lot of negative space, the elements on the screen do not appear cluttered. They must not be cluttered because students will need to feel like the learning platform is organised.

Colour – The colours we used are the brand colours of Makukhanye Edu-Care in combination with some basic colours. Blue works because it is a primary colour that places itself in the foreground of the composition. According to various colour theories, the specific version of blue that we chose represents trustworthiness, modernity depth and intelligence.

Shape & Form – We decided to use a mixture of squares and rectangles because they are easily recognisable shapes. This makes the design simple, easy to understand and conceptually accessible. Most of the icons have rounded edges to make them more aesthetically pleasing, and rounded edges also help students distinguish between different icons.

Unity – Unity is achieved by sticking to a modern simple style of design.

Visual Hierarchy – Visual hierarchy was created using a mixture of colour, shape, form, balance and proportions.

We started our design work with a competitive analysis. The following design approaches helped to understand possible design approaches: user-centred design, participatory design and cognitive modelling. We believe that limiting ourselves to online learning

platforms would hinder our ability to brainstorm design solutions. The competitive analysis aided us to come up with an effective and modern design. Below is an example of an existing e-learning student portal dashboard that we used with the students to understand their eye tracking and see the ability to navigate through the portal.

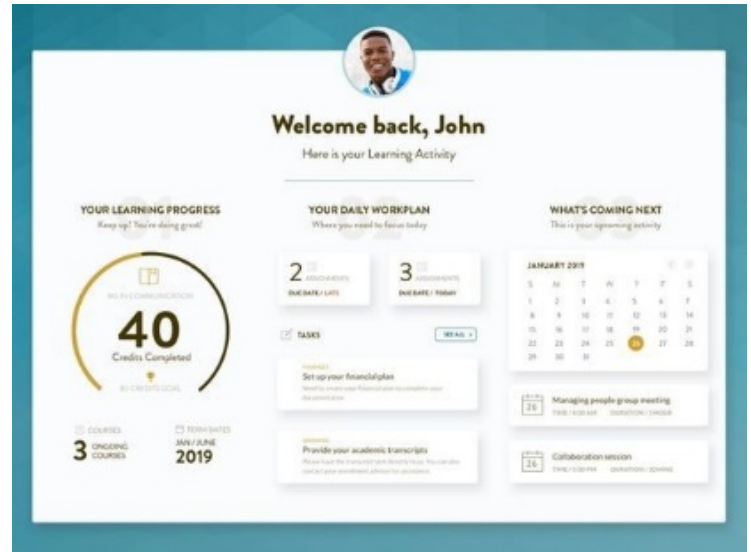


Figure 3: Dribbble. E-learning Portal: Student dashboard, via Pinterest

(<https://za.pinterest.com/pin/1082623197948979/>)

This is a goal oriented UX design. There are visual cues that promote progress. There is an effective use of repetition to create a visual hierarchy. We believe that the visual hierarchy will create a psychological effect on the user. When you open the app, the first thing you see because of its visual weight made by its being a warm primary color, is your current progress. Tellingly, seeing your progress will motivate you to see what more you can do to move closer to your goals. You will look towards the left and see a daily work plan with your task checklist. On the far right, it shows the calendar with things that are yet to come.

14 FUTURE WORK

In preparation for this submission, we acknowledge that the work is partially done. We wanted to add a careers portal where the students will be able to get career advice and inform them about the careers that are in demand. This would give the student a clear picture and understanding of the career they will choose and also help them make a good career choice. We also wanted to add a girls' talk section which would focus on woman empowerment and trying to get the young ladies to believe that they can make a huge difference in this world. The students we were co-designing with believed that the careers portal and the girl talk should be the second phase of the development process. They want to see the content first, which is a burning issue for them.

The testing phase is supposed to be carried out during February 2023. The reason for this decision is that students only return to school on the 18th of January 2023. We engaged with the students

and agreed that February was a realistic date. This gave us time to get the curriculum for the content creators and also make sure that the content is up to standard.

15 CONCLUSION

In summary, this project aimed to address the limited access to quality education in Eastern Cape by developing a digital learning portal for high school students in rural areas. The project used a combination of a participatory design approach and user-centred design methods to engage with students and gather their needs and preferences. The resulting website includes features such as digital libraries, online teaching, academic support, simplified notes and easily understandable video lessons, messaging with tutors and practice questions. The design process included user interface design and prototyping tools and focused on delivering content that was accessible and relevant to the target audience. The testing phase results showed that the portal effectively addressed the challenges faced by students and provided valuable support for their education. Going forward, future evaluations and monitoring will be crucial to continuously improve the system and ensure its effectiveness in delivering quality education.

16 ACKNOWLEDGMENTS

Many thanks to the students of Vulamazibuko High School and Masixole Secondary School for participating in our primary research. We want to thank Walter Sisulu University, Nordi CHI and Professor Christian Sturm for their assistance in making this project a success. We want to extend special thanks to Professor Nobert Jere and Ms Nosipho Mavuso, who are our academic advisors and mentors.

REFERENCES

- [1] Echazarra Alfonso and Radinger Thomas. 2019. Learning in rural schools: Insights from PISA, TALIS and the literature. OECD Education Working Papers, No. 196. OECD Publishing, Paris. Retrieved from <https://doi.org/10.1787/8b1a5cb9-en>.
- [2] United Nations Educational Scientific and Cultural Organization 2017. 2017/8 Accountability in education: Meeting our commitments. Paris, France: Author. Retrieved from <https://apo.org.au/sites/default/files/resource-files/2018-04/apo-nid1433912.pdf>
- [3] Chavez Filiberto.V. 2016. Spatial mapping for improved intra-urban education planning. Paris, France: UNESCO. Online Report. Retrieved from <https://core.ac.uk/download/pdf/143615053.pdf>
- [4] Daniel Luke. 2021. 99% of public schools in the eastern cape aren't proper fenced and pit latrines abound, Business Insider SA Education report focusing on the Eastern Cape, 2016, Statistics South Africa. Online Report. Retrieved from 99% of public schools in the Eastern Cape aren't properly fenced – and pit latrines abound | Business Insider
- [5] Education report focusing on the Eastern Cape. 2016. Statistics South Africa. Online Report. Retrieved from Education report focusing on the Eastern Cape | Statistics South Africa (statssa.gov.za)
- [6] Ellis Estelle. 2022. Eastern Cape: Class of 2021 significantly improve pass rate despite losing 58% of teaching time in 2020, Daily Maverick. Online Report. Retrieved from Eastern Cape: Class of 2021 ups pass rate despite losin... (daily-maverick.co.za)
- [7] Grant Donald. 2013. Background to the national quintile system, Western Cape Education Department. Online Report. Retrieved from WCED: Media release (westerncape.gov.za)
- [8] Nowicki Lucas. 2022. Court to get education department to deliver textbooks, Groundup. Online Report. Retrieved from Court to get education department to deliver textbook | City Press (news24.com)
- [9] Zama Zanele. 2022. Importance of women studying STEM subjects in schools, 702 radio. Retrieved from 'SA only has 7% of females in science, technology, engineering and mathematics' (702.co.za)
- [10] Sizani Mkhusele. 2022. Overcrowded schools accuse government of breaking promises GroundUp, Online Report Retrieved from Overcrowded schools accuse government of breaking promises | GroundUp
- [11] Glatstein, Jerney 2019. Formal visual analysis: The elements & principles of composition. The Kennedy Center. Online Report Retrieved from Formal Visual Analysis: The Elements & Principles of Composition (kennedy-center.org)
- [12] Loggenberg Camel. 2012. EC learners sit idle waiting to be taught, Eyewitness News. Online Report. Retrieved from EC learners sit idle, waiting to be taught (ewn.co.za).
- [13] Timothy Neate, Aikaterini Bourazeri, Abi Roper, Simone Stumpf and Stephanie Wilson. 2019. Co-Created Personas: Engaging and Empowering Users with Diverse Needs Within the Design Process. CHI'19: Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (May 2019), 1-12. DOI: <https://doi.org/10.1145/3290605.3300880>
- [14] Artemis Skartatidou, Carol Iglesias Otero. 2021. Geographic Citezen Science Design: No one left behind. Chapter3 Design Approaches and human-computer interaction methods to support user involvement in citizen science. DOI: <https://doi.org/10.2307/j.ctv15d8174.11>
- [15] Asvhns Pierre Du Plessis. 2014. Problems and Complexities in Rural Schools: Challenges of Education and Social Development. Mediterranean Journal of Social Sciences. Volume 5. DOI - 10.5901/mjss.2014.v5n20p1109
- [16] Aimee Howley, Megan Rhodes, and Jimmie Beall. 2009. Challenges Facing Rural Schools: implications for Gifted Students. Journal for the Education of the Gifted. Vol. 32, No. 4, 2009, pp. 515–536.
- [17] Kabatangare, Tumuhairwe. 2021. Data literacy integration into development agenda. A catalyst to achieving the Sustainable Development Goals (SDGs). IASSIST Quarterly. Volume 45. DOI - 10.29173/iq1003
- [18] Brovarone, Elisabetta and Cotella, Giancarlo. 2020. Improving Rural Accessibility: A Multilayer Approach. Sustainability Journal. 17(7):2876. DOI - 10.3390/su12072876
- [19] Calderon, Cesar; Cantu, Catalina. 2021. *The Impact of Digital Infrastructure on African Development*. Policy Research Working Paper; No. 9853. World Bank, Washington, DC. ©World Bank. <https://openknowledge.worldbank.org/handle/10986/36629> License: CC BY 3.0 IGO."
- [20] Lucas Nowicki. 2022. 12 years after a court ordered it, government plans to build a special needs school. Online Report. Available at: <https://www.groundup.org.za/article/12-years-after-a-court-ordered-it-to-government-promises-to-build-special-needs-school/>